Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A car is moving away from the base of a 30 m high tower. The angle of elevation of the top of the tower from the car at an instant when the car is $10\sqrt{3}$ m away from the base is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

Q2. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground inclined at 60°. Assuming no slack, the length of the string is: [1]

(A) $40\sqrt{3}$ m (B) $30\sqrt{3}$ m (C) $20\sqrt{3}$ m (D) 120 m

Q3. If the height of a vertical pole is $\sqrt{3}$ times the length of its shadow on the ground, then the angle of elevation of the sun is: [1]

(A) 30° (B) 45° (C) 60° (D) 75°

Q4. An observer 1.5 m tall is 28.5 m away from a tower 30 m high. The angle of elevation of the top of the tower from the eye of the observer is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

Q5. **Assertion (A):** The angle of depression of an object viewed into a valley from a cliff is numerically equal to the angle of elevation of the cliff viewed from that object.
Reason (R): The line of sight acts as a transversal line intersecting two parallel horizontal planes, making alternate interior angles equal. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

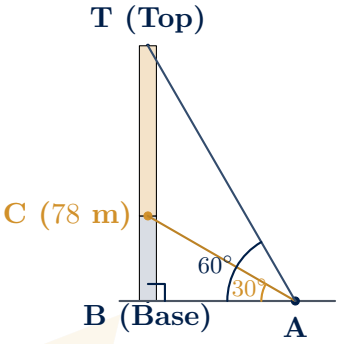
Q6. An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45°. What is the height of the chimney? [2]

Q7. From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30° and 45°, respectively. If the bridge is at a height of 3 m from the banks, find the width of the river. [3]

Q8. A 1.2 m tall girl spots a balloon moving with the wind in a horizontal line at a constant height of 88.2 m from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60°. After some time, the angle of elevation reduces to 30°. Find the distance travelled by the balloon during the interval. [3]

Q9. An aeroplane flying horizontally at a constant height of $1500\sqrt{3}$ m above the ground is observed at an elevation of 60°. After 15 seconds, the angle of elevation is observed to be 30°. Find the speed of the aeroplane in km/h. [3]

Q10. CASE STUDY: The Iconic Pitampura TV Tower (Doordarshan Delhi)



The TV Tower located at Pitampura, Delhi was constructed in 1988 and is one of the tallest architectural towers in Northern India. The tower stands vertically on level ground.
A surveying student standing at point A on the ground observes the top of the tower (T) at an angle of elevation of $60°$. The tower has an observation and technical gallery located at point C, which is situated at a vertical height of 78 m directly above the base of the tower (B). The angle of elevation of point C from point A is found to be $30°$.

Based on the above mathematical scenario, answer the following questions:

(i) Find the horizontal distance between the observation point A and the base of the TV tower (B). [1]

(ii) Find the direct line-of-sight distance from point A to the observation gallery at point C. [1]

(iii) Calculate the total height of the Pitampura TV Tower (BT). [2]

— OR —

Find the vertical distance between the observation gallery C and the top of the tower T. [2]

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